

Mathematical Problem Solving I

Logic, Invariants, Colouring and Induction



Overview

1 Competitive Programming

2 Logic

3 Invariants

4 Colouring Arguments

5 Induction and Monovariants



Competitive Programming

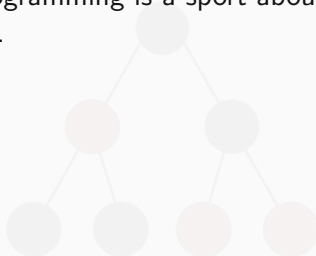


What Is Competitive Programming?



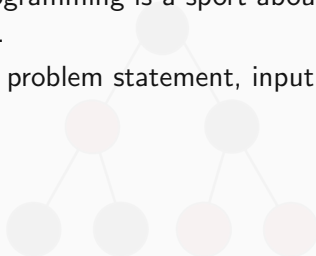
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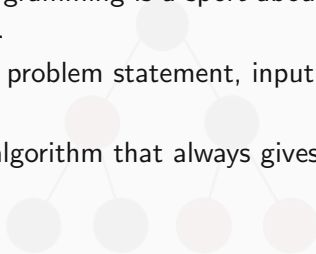
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- The challenge is not just solving it once. It is solving every valid case quickly enough.

Why Start With Mathematical Puzzles?



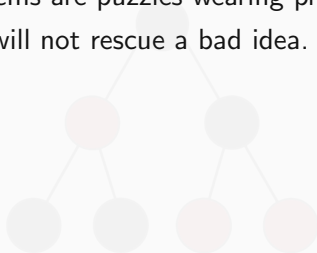
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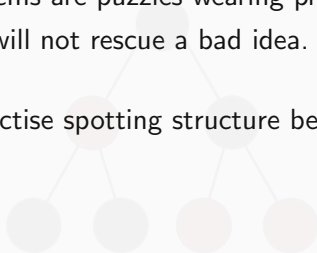
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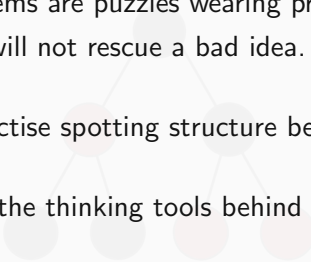
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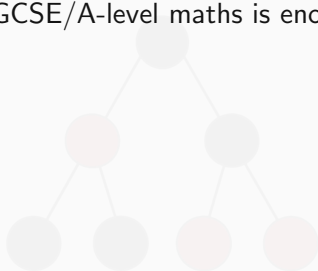
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- Many CP problems are puzzles wearing programming clothes.
 - The computer will not rescue a bad idea. It will just run it faster.
 - We want to practise spotting structure before worrying about syntax.
 - Today is about the thinking tools behind many algorithmic solutions.
 - Once the idea is clear, implementation becomes much less mysterious.

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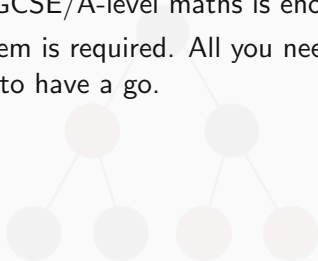
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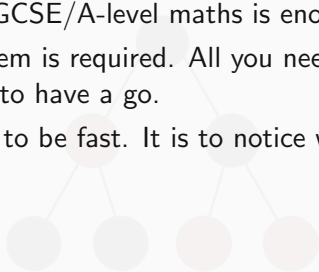
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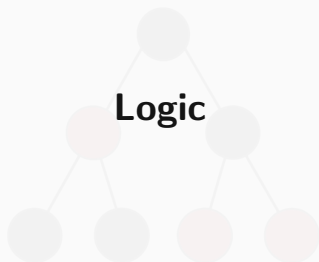


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- If you already have a stronger foundation, we have left some harder questions for you to tackle.

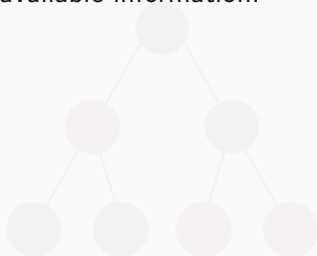


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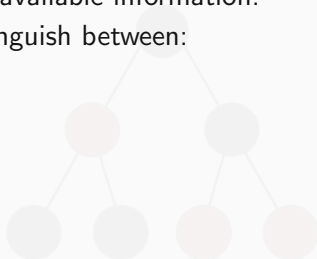
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- Logic is the study of **reasoning**: how conclusions follow from assumptions or available information.



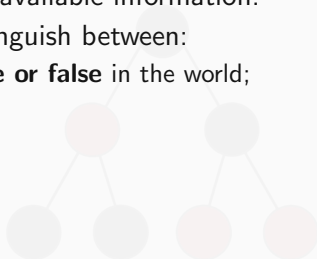
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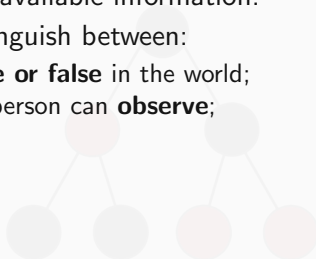
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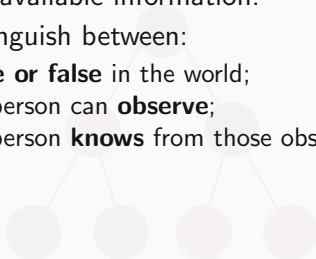
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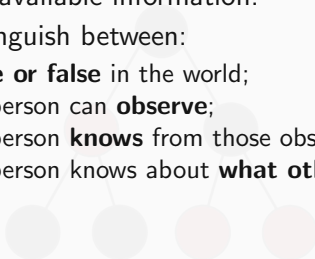
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- A central question is:

"Given what I know, and what I know about what others know, what must be true?"

Worked Example: Two Doors

You face two doors. One leads out; one does not.

One guard always tells the truth. The other always lies. You do not know which guard is which.

You may ask one guard one yes/no question. How do you find the safe door?

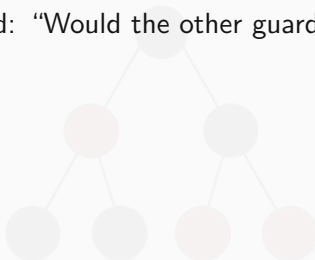
Logic

Solution: Two Doors



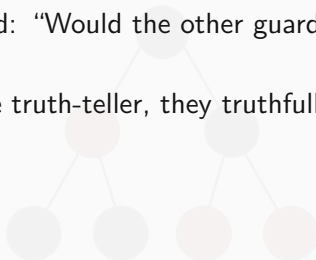
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→ Ask either guard: “Would the other guard say this door is safe?”

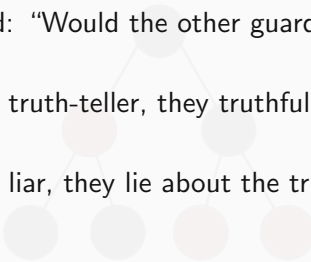


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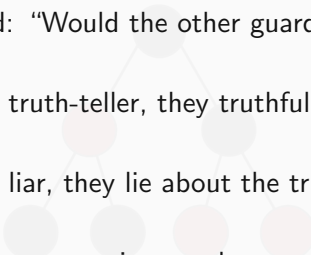
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 - If you asked the liar, they lie about the truth-teller’s correct answer.
 - Either way, the answer points to the unsafe door. Choose the other one.

Student Exercises

- Basic: Same Birthday Month.
- Easy: Glove Selection.
- Medium: Two Doors.
- Hard: Islander Statements.
- Very Hard: Counterfeit Coin.

Invariants



What is an Invariants?

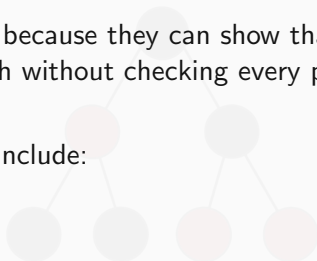


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An invariant is a quantity or property that stays unchanged after every legal move.

Invariants are useful because they can show that a certain position is impossible to reach without checking every possible sequence of moves.

Common invariants include:



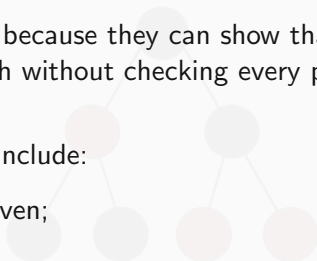
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- number of connected components;
- colour balance;
- XOR or binary properties.

Worked Example: Last Ball

A bag contains black and white balls.

Repeatedly remove two balls:

- same colours: put back one black ball;
- different colours: put back one white ball.

Can we predict the colour of the last ball?

Invariants

Solution: Last Ball



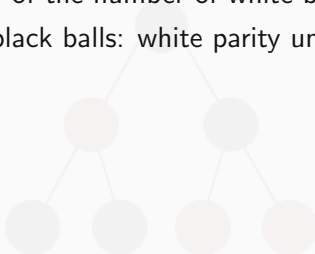
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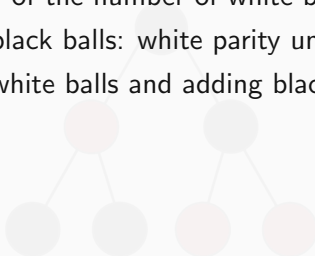
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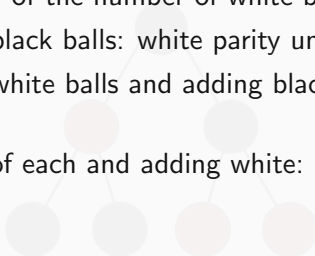
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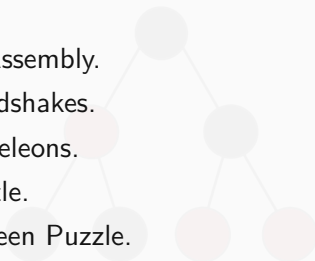
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- Removing one of each and adding white: white count unchanged.
- The parity of the number of white balls is invariant.
- The last ball is white if the initial white count was odd, and black if it was even.

Student Exercises

- Basic: Jigsaw Assembly.
- Easy: Odd Handshakes.
- Medium: Chameleons.
- Hard: MU Puzzle.
- Very Hard: Fifteen Puzzle.



Colouring Arguments



What are Colouring Arguments?



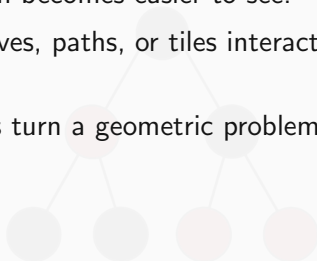
What are Colouring Arguments?

A **colouring argument** assigns colours to parts of a problem so that a hidden pattern becomes easier to see.

It is useful when moves, paths, or tiles interact with the colours in a predictable way.

Colouring arguments turn a geometric problem into a counting or parity problem.

Examples:



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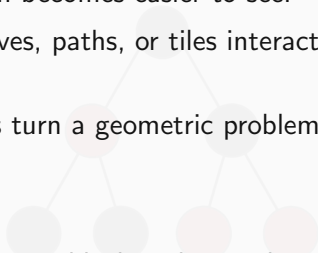
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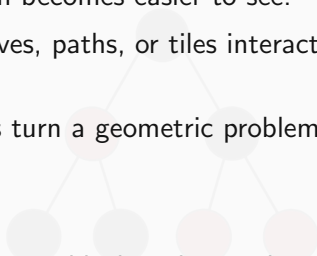
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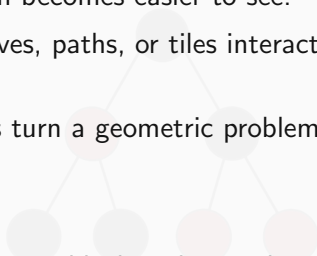
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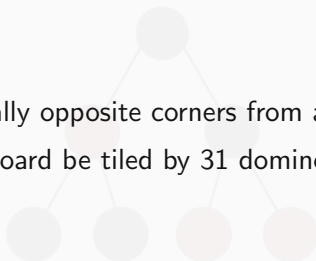
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- a knight always moves to a square of the opposite colour;
- a path through adjacent grid squares alternates between colours.



Worked Example: Deficient Chessboard

Remove two diagonally opposite corners from a chessboard.
Can the remaining board be tiled by 31 dominoes?

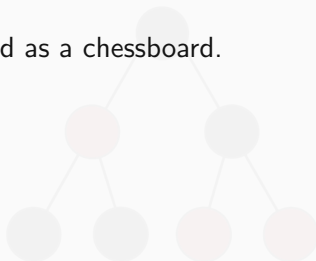


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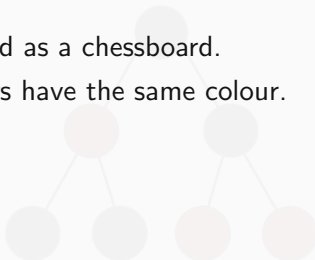
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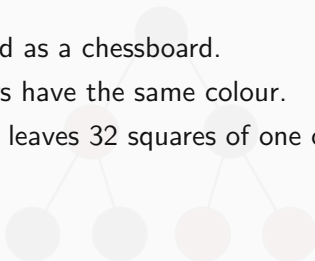
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- Colour the board as a chessboard.
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- Removing them leaves 32 squares of one colour and 30 of the other.



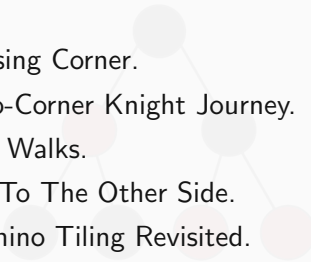
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- Every domino covers exactly one of each colour.
- A tiling would need equal colour counts, so it is impossible.

Student Exercises: Colouring

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- Basic: One Missing Corner.
 - Easy: Corner-To-Corner Knight Journey.
 - Medium: Board Walks.
 - Hard: Jumping To The Other Side.
 - Very Hard: Domino Tiling Revisited.

Induction and Monovariants



What is Induction?



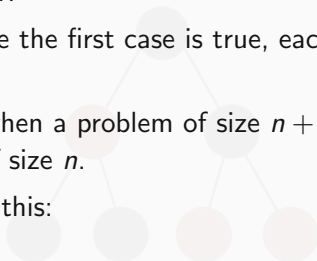
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Induction is a method for proving that a statement is true for every positive integer.

The idea is that once the first case is true, each case forces the next one to be true.

Induction is useful when a problem of size $n + 1$ can be related to the same problem of size n .

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- **Induction step:** assume the statement is true for some n , and use this to prove it is true for $n + 1$.

Monovariants

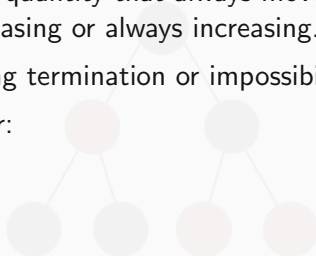


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It is useful for proving termination or impossibility.

What to look out for:



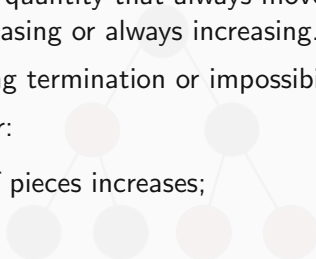
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What to look out for:

- total number of pieces increases;
- distance to a target decreases;
- perimeter never increases;
- number of bad pairs decreases.

Worked Example: Chocolate Bar

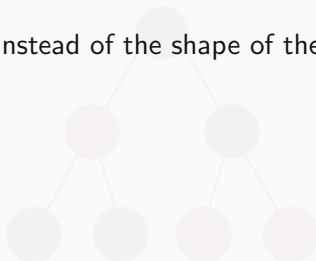
An $n \times m$ chocolate bar must be broken into unit squares.
Each move breaks exactly one existing piece in a straight line.
What is the minimum number of breaks?

Solution: Chocolate Bar



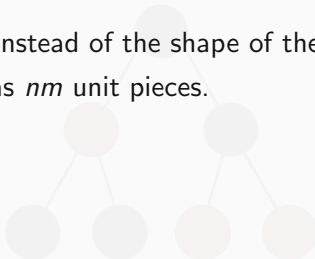
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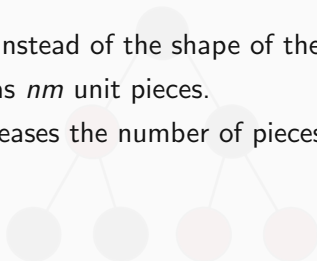
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- Track progress instead of the shape of the pieces.
- The final bar has nm unit pieces.

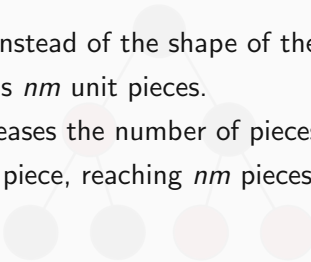


Solution: Chocolate Bar

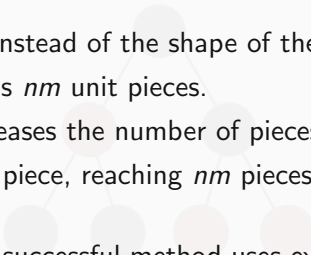
- Track progress instead of the shape of the pieces.
- The final bar has nm unit pieces.
- Each break increases the number of pieces by exactly 1.



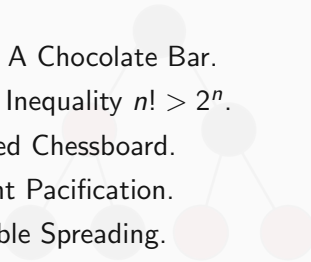
Solution: Chocolate Bar

- 
- Track progress instead of the shape of the pieces.
 - The final bar has nm unit pieces.
 - Each break increases the number of pieces by exactly 1.
 - Starting from 1 piece, reaching nm pieces needs $nm - 1$ increases.

Solution: Chocolate Bar

- 
- Track progress instead of the shape of the pieces.
 - The final bar has nm unit pieces.
 - Each break increases the number of pieces by exactly 1.
 - Starting from 1 piece, reaching nm pieces needs $nm - 1$ increases.
 - Therefore every successful method uses exactly $nm - 1$ breaks.

Student Exercises: Induction and Monovariants

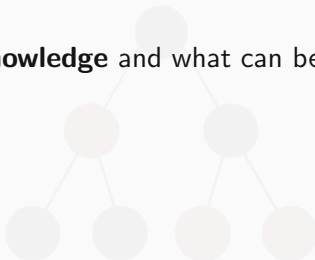
- 
- Basic: Breaking A Chocolate Bar.
 - Easy: Induction Inequality $n! > 2^n$.
 - Medium: Infected Chessboard.
 - Hard: Parliament Pacification.
 - Very Hard: Pebble Spreading.

Wrap-Up



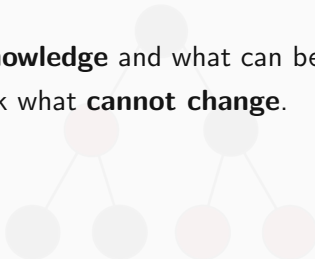
Wrap-Up

→ **Logic** tracks **knowledge** and what can be inferred.



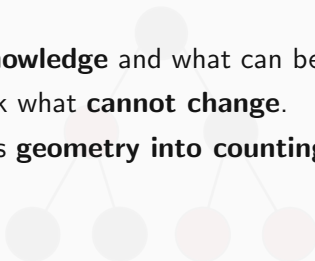
Wrap-Up

- **Logic** tracks **knowledge** and what can be inferred.
- **Invariants** track what **cannot change**.

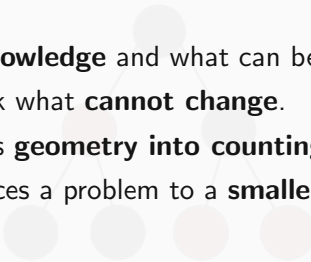


Wrap-Up

- **Logic** tracks **knowledge** and what can be inferred.
- **Invariants** track what **cannot change**.
- **Colouring** turns **geometry into counting**.



Wrap-Up

- 
- **Logic** tracks **knowledge** and what can be inferred.
 - **Invariants** track what **cannot change**.
 - **Colouring** turns **geometry into counting**.
 - **Induction** reduces a problem to a **smaller case**.

Wrap-Up

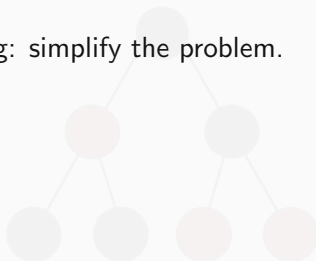
- **Logic** tracks **knowledge** and what can be inferred.
- **Invariants** track what **cannot change**.
- **Colouring** turns **geometry into counting**.
- **Induction** reduces a problem to a **smaller case**.
- **Monovariants** track **progress** and can reveal **impossibility**.

What's more?



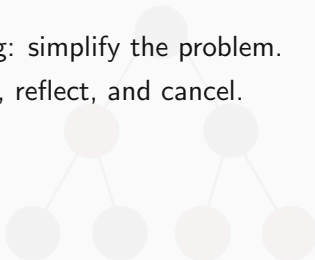
What's more?

→ Wishful thinking: simplify the problem.



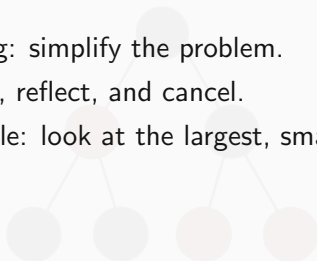
What's more?

- Wishful thinking: simplify the problem.
- Symmetry: pair, reflect, and cancel.

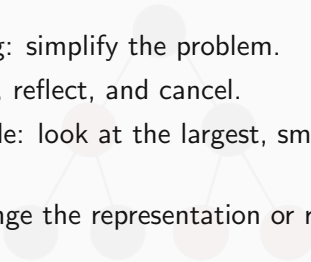


What's more?

- Wishful thinking: simplify the problem.
- Symmetry: pair, reflect, and cancel.
- Extreme principle: look at the largest, smallest, or boundary case.



What's more?

- 
- Wishful thinking: simplify the problem.
 - Symmetry: pair, reflect, and cancel.
 - Extreme principle: look at the largest, smallest, or boundary case.
 - Reduction: change the representation or reduce to a familiar problem.

Thanks for coming!

See ya next week!

I hope you enjoyed the workshop and found some
useful new ways to approach problems.